GraphARC: A Comprehensive Benchmark for Graph-Based Abstract Reasoning

ETH zürich

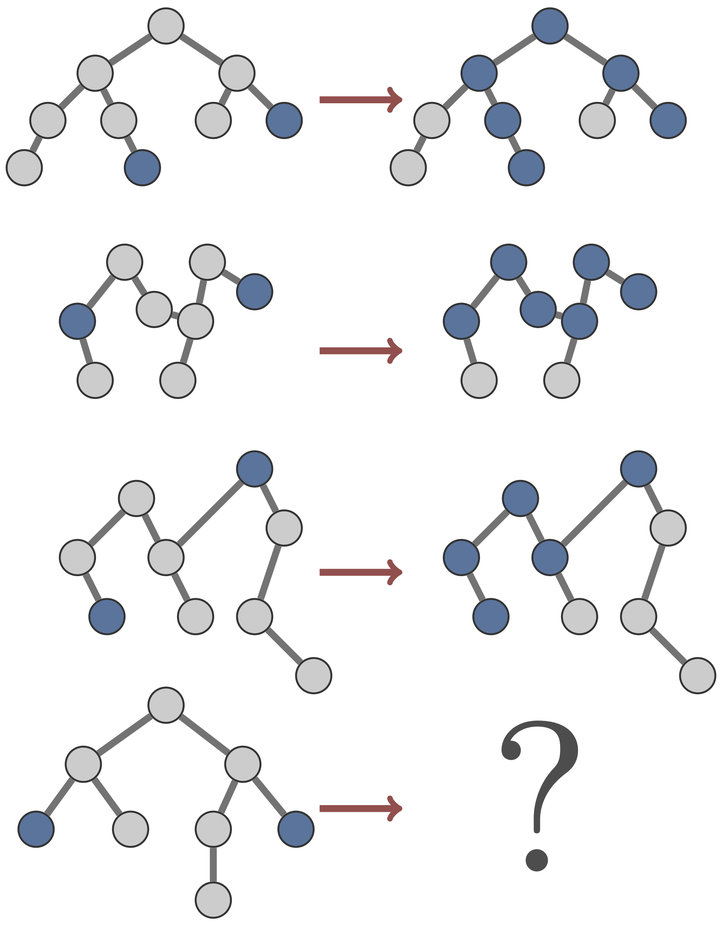
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Introduction & Motivation

Relational reasoning—the ability to understand and manipulate relationships between entities—is fundamental to intelligence. Most abstract reasoning benchmarks, however, center on fixed-size grids and spatial patterns. Inspired by ARC, we propose GraphARC to benchmark few-shot abstract reasoning on graphs.

Task Setup

Infer a transformation from a set of input–output graph pairs and apply it to a new test graph.



Why GraphARC?

- **Structure without layout:** graphs represent objects and their relationships directly, independent of spatial arrangement.
- **Scalable generation:** each transformation yields a virtually unlimited supply of instances across graph families and sizes.
- **Systematic generalization:** GraphARC tests whether rules inferred from a few examples transfer to new graph structures and scales.
- **Unified graph reasoning:** one framework combines node classification, link prediction, and graph generation.

Graph Transformations

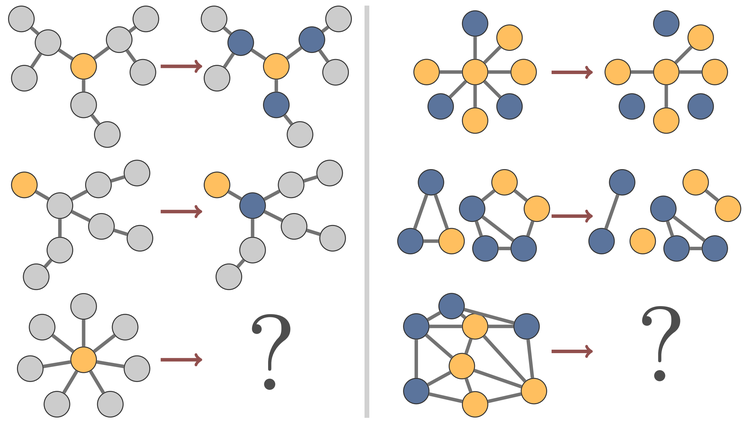
Color-based tasks *(preserve structure)*

- colorDegree1/2/3, colorMax/MinDegree, colorNeighbors
- colorPath, colorComponents, bipartitionCompletion

Structure modification *(alter topology)*

- addHub, edgeToNode, removeDegree1/2/3
- blueSubgraph, complementGraph, mergeAtBlue

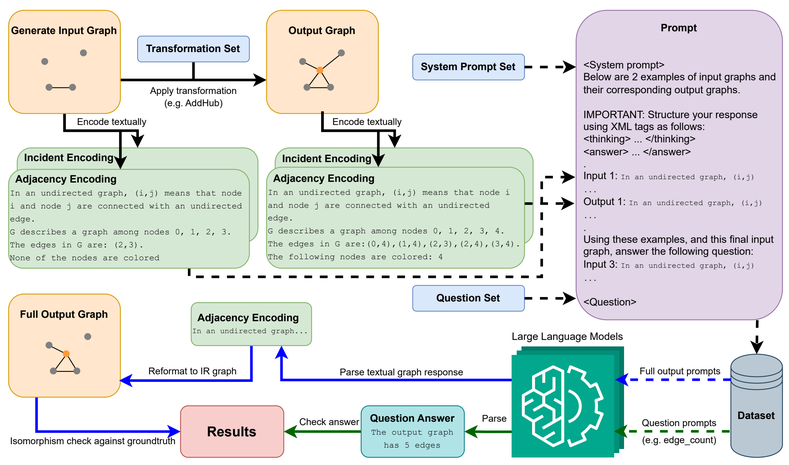
Additional Example Tasks



Evaluation Paradigms for LLMs

Full-output path: Complete graph generation → tests both comprehension and execution.

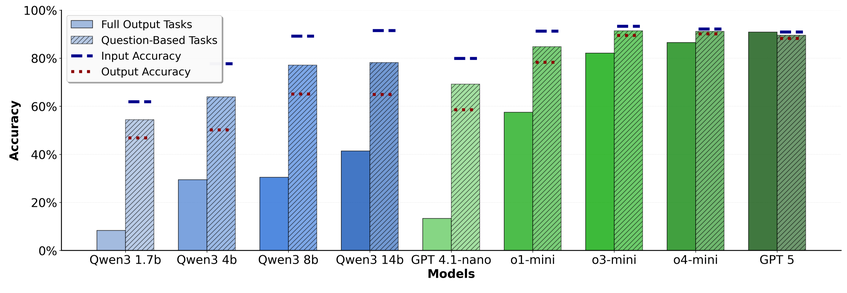
Question-based path: Targeted property questions → isolates comprehension from execution.



Dataset & Evaluation

- 125,000+ model evaluations.
- Qwen3 (1.7B–14B), GPT-4.1-nano, o1/o3/o4-mini, GPT-5.
- Graph sizes: 5–250 nodes.
- Variations: encodings, system prompts, graph families.
- Graph types: ER, Watts–Strogatz, trees, bipartite.

Overall Model Performance



Key Insights

1. Comprehension–execution gap:
Models understand graph properties but struggle to **generate consistent transformations**.

- Qwen3 1.7B: **6.55× gap** (54.4% vs. 8.3%).
- Qwen3 8B: **2.36× gap** (72.2% vs. 30.5%).

2. Scaling barriers:
Mid-tier models collapse at 50–100 nodes; local operations can perform **worse** than global ones.

- o1-mini: 88% at 10 nodes → **18%** at 250 nodes.
- GPT-5 maintains **91%** even at 250 nodes.

3. Paradox of capability
Some more capable models apply transformations even when they are **not asked** to do so.

Task-Specific Performance

Task	Qwen3 14B	DeepSeek-R1 LLaMA-8b	GPT 4.1-nano	o1 mini	o3 mini	GPT 5	Avg
removeDegree3	0.02	0.00	0.00	0.00	0.25	**0.75**	0.15
colorEquidistant	0.00	0.00	0.00	0.38	0.50	**0.75**	0.23
colorDegree3	0.29	0.10	0.00	0.00	**0.67**	0.58	0.24
colorDistanceAtLeast2	0.00	0.00	0.00	0.08	0.83	**0.92**	0.26
removeDegree2	0.00	0.00	0.00	0.42	0.75	**1.00**	0.31
colorMaxDegree	0.52	0.04	0.25	0.25	0.33	**0.67**	0.33
edgeToNode	0.25	0.00	0.00	0.50	0.88	**1.00**	0.38
bipartitionCompletion	0.00	0.00	0.00	0.75	**1.00**	**1.00**	0.39
mergeAtBlue	0.00	0.00	0.00	0.75	**1.00**	**1.00**	0.39
colorDegree2	0.42	0.06	0.04	0.42	0.92	**1.00**	0.41
complementGraph	0.19	0.16	0.05	0.69	**1.00**	**1.00**	0.46
colorMinDegree	0.58	0.29	0.12	0.67	0.75	**0.83**	0.47
colorInternal	0.44	0.20	0.25	0.62	**1.00**	**1.00**	0.54
colorPath	0.41	0.09	0.31	0.88	**1.00**	**1.00**	0.55
removeDegree1	0.41	0.34	0.19	0.75	**1.00**	**1.00**	0.55
colorComponents	0.62	0.44	0.19	**1.00**	0.75	**1.00**	0.58
colorNeighbors	0.88	0.14	0.44	**0.94**	**0.94**	0.88	0.63
addHub	0.66	0.50	0.20	0.75	**1.00**	**1.00**	0.64
colorDegree1	0.88	0.45	0.27	0.88	**1.00**	**1.00**	0.67
Average	0.37	0.17	0.12	0.58	0.81	0.91	0.44

Conclusion

GraphARC is a **scalable benchmark for graph reasoning**:

- 21 color and structure-based transformations
- Graph sizes 5–250 enable controlled scaling tests.
- Dual evaluation covers outputs and graph-property questions.
- Future work will evaluate **graph foundation models**.

Code: github.com/sakupeltonen/graph-arc

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